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UBoot64 v2



Boot menu for C64 Ultimate enabled devices — v2 (Oscar64 rebuild)

Contents

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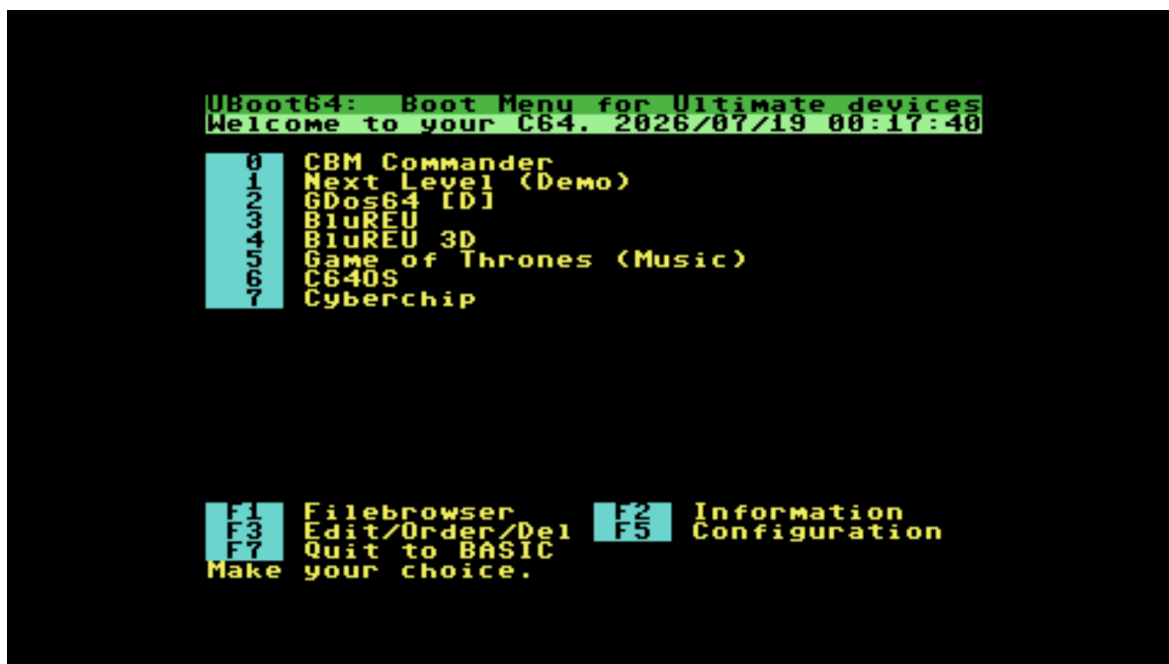
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Version history and download

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Version 4.0.0 - (in development):

- Directory block sizes in IEC mode are no longer shown modulo 256 (a 300-block file showed as 44).
- IEC directory listings: locked files (PRG<) and file types UBoot64 does not know no longer replace the disk name at the top of the listing; they show as normal entries.
- Slot editor: an earlier change is no longer lost when a later action in the same session changes nothing.
- Configuration: changes are no longer lost when the colour editor is left without changes; **DEL** (undo) in the colour editor now only undoes the colour changes, not other unsaved configuration changes.
- NTP time sync: the date is now correct in leap years from March on (February always had 28 days).
- NTP time sync: three NTP servers, tried in order until one answers (as in DMBoot 128 v5). NTP sync is now off by default for new configurations, since current Ultimate firmware sets the clock itself; existing configurations keep their setting.
- Start-up messages: a third option, **Show messages + wait**, keeps the start-up feedback on screen until a key is pressed.
- "IDs needing manual power switching" now reports "Yes" when any such device is on the bus, not only when it is the last one checked.
- REU preload: the path of the REU image is now stored in the slot. Before, a slot loaded the REU file from the directory of its drive A image. Existing slots are converted automatically: by

the upgrade tools, or at the first boot of the slot.

- Firmware 3.15+ SoftIEC drive (IEC mode):
 - Slots made on the SoftIEC drive now use UBoot64's own root partition and the drive's real host path, so they keep working whichever partition or directory you browsed in, and no dirtrace is needed.
 - **M** (mount and run) now also works for a disk image entered on the SoftIEC drive in IEC mode: the slot mounts that image on drive A and runs the chosen file.
 - REU images chosen in IEC mode now get their real Ultimate path, so they can be preloaded.
- **F4** (partitions) on a drive without partition support now says so, instead of listing its files as partitions.
- The USR file type showed as "URS".
- A new REU or mount-only slot gets the file name as its default name (it started empty, which hid the slot in the menu); an emptied name falls back to it.
- The file browser's side menu shows **M** (run mount) in IEC mode inside a disk image.
- Messages in the file browser no longer leave the last letter of the file type column on screen.
- Internal: fixed functions returning pointers to temporary buffers, and made the REU transfers of the slot and directory data safe against compiler reordering.
- Major version 4: the configuration and slot file format changed again (format v4, a settings file per slot). Version 4.0.0 converts the files of all earlier versions itself (see below).
- Old configuration and slot files (v1, v2 and v3) are converted at start-up, after asking, with copies of the old files kept as `dmbcfg.v1/dmbst.v1` (or `.v2`, `.v3`). The file format is now v4 (a settings file per slot). The separate upgrade tools are no longer needed (they still come with this release, for v1 and v2).
- Start-up: the storage device for the configuration and slot files is now found from one listing of the Ultimate's root directory, so any SD or USB device counts (before: only `/sd/`, `/usb0/`, `/usb1/` and `/usb2/` were tried). The search order is still SD first, then USB; the upgrade tools search the same way.
- Program settings files: when a slot boots, UBoot64 applies the program's own Ultimate settings file (`<image or program>.cfg`, else `.usr`), as the Ultimate's own file browser does (firmware 3.15+), and at every start a baseline file `uboot64/uboot64.cfg`. A slot can also get its own settings file: **S** in the file browser. Switch it all off with **C** in **F5**. See [Program settings files](#).
- Loose `.prg` files can now be added to a slot from the file browser's UCI mode (**ENTER**): UBoot64 loads them through the REU, without a drive or disk image. See [Programs from UCI mode](#).
- **F2** (information) shows the Ultimate's product name (for example "Ultimate 64 Elite"), or "unknown" if the firmware doesn't report it.
- Internal: the Ultimate Command Interface library is now [ultimate-uci-oscar64](#) (v1.3.0), included as a git submodule. It brings a fix for a hang at start-up found by Christian Gleissner: a command sent while the interface was still busy could leave it waiting for ever.

Version 3.0.1 - 20260913:

- Fixed a regression against Ultimate firmware 3.15a: the "SoftIEC root partition" auto-provisioning (config menu, **F8**) could misreport the reserved partition as already in use by something else and refuse to set itself up, because firmware 3.15a changed what the partition

listing's name field contains. No user-visible change on firmware 3.15 or earlier.

- Fixed a boot-time gap: a menu slot recording the "SoftIEC root partition" would select it via the classic CP command without first making sure it actually still existed. Since that partition only lives in the Ultimate's memory (not saved to flash unless you do so yourself), booting straight into such a slot after a power cycle -- without ever browsing into IEC mode first -- could send that select command against a partition that was no longer there. The slot now re-creates it (harmlessly, if it's already present) before selecting it.
- Fixed a related bug uncovered by the above: a menu slot's recorded directory path, when rebuilt for a SoftIEC or VICE device, used a single leading slash (`cd:/path`). Ultimate firmware treats a single leading slash as relative to wherever the drive's directory position currently is, not as absolute from the drive's root -- only a double slash (`cd://path`) is. This had quietly relied on the drive always still being positioned at its root the first time a slot ran; selecting a specific SoftIEC partition first (see above) no longer guarantees that, so affected slots could fail to find their file. Slots now save with the correct double-slash form; existing slots pick this up automatically the next time they're saved (re-edited or re-created).
- Fixed a display glitch where the "partition already in use" message (see above) could overflow into the side menu panel and show a stray character, instead of staying inside its own message box.
- Fixed "go up one directory" (**DEL**) on real IEC hardware that reports CMD-HD-style partitions (e.g. SD2IEC) sending an invalid command that the drive silently rejected, while UBoot64's own idea of the current directory moved anyway -- so the display and the drive disagreed on where you were, showing stale or jumbled directory content on the next move. Only affected non-Ultimate IEC devices; the Ultimate's own SoftIEC drive was unaffected.
- Fixed the "SoftIEC root partition" conflict check occasionally misfiring even against its own, correctly-created partition, due to an internal text-encoding inconsistency between where the partition is created and where it's checked.
- Removed the "delete partition from device" prompt when turning the "SoftIEC root partition" option off (confirmed on real hardware that the underlying firmware command doesn't actually remove it). Since the partition was never saved to flash in the first place, a power cycle already clears it -- there was nothing this prompt could reliably do.

Version 3.0.0 - 20260907:

- Compatibility with Ultimate firmware 3.15+: UCI now auto-enables itself from the cartridge (no need to turn it on in the Ultimate menu beforehand), and the classic-IEC "go up one directory" command adapts to the rewritten SoftIEC DOS parser automatically, with no change in behaviour on older firmware.
- SoftIEC partition support (firmware 3.15+ adds CMD-HD-style partitions to SoftIEC, and this works the same way on any IEC device that supports partitions, e.g. CMD-HD or SD2IEC): press **F4** while browsing in IEC mode to show a list of the device's partitions; select one to browse into it. Pressing **DEL** at a partition's own root returns to this list rather than trying to leave the device entirely. An opt-in "SoftIEC root partition" toggle (config menu, **F8**) auto-creates a partition exposing the whole filesystem at root the next time you switch to IEC mode (**F3**), without ever touching a partition you've configured yourself — turning it back off offers to remove that partition from the device too. This partition lives only in the Ultimate's memory until you save it via the Ultimate's own UI — see [F5: Configuration](#) for details. Menu slots can

now record and restore a specific partition at boot.

- Slot/config file format changed to add partition support — see [Upgrading from an earlier version](#). Major version bumped to 3.0.0 to reflect this.

Version 2.2.0 - 20260816:

- USB port auto-reroute: if a slot's disk image or REU file was saved against one USB port (USB0/1/2) and the stick has since been moved to another, UBoot64 now detects this at boot time, automatically retries on the other USB ports, and reroutes to it for that boot if found — no need to re-edit the slot. Checks the actual file, not just the folder, so it correctly skips a USB stick that happens to share a folder name but not the file itself. See [USB port auto-reroute](#).
- If the stick isn't found on any USB port, boot now stops with a clear "insert USB stick" message and lets you retry after reinserting it, instead of a generic mount error — press F7 to give up and return to BASIC instead.

Version 2.1.1 - 20260719:

- Configuration and slot files can now live on an SD card as well as USB storage. On startup, UBoot64 looks in this order: SD, USB0, USB1, USB2, and uses the first location where files already exist; a fresh install creates them on the first of those that's actually present, in the same order. Previously only a single USB drive was checked, and an SD-only setup would fail to start.
- The v1-to-v2 upgrade tool (uboot_upd12.prg) uses the same search order, so an upgrade always lands on the same device the cartridge will look for it on.

Version 2.1.0 - 20260719:

- Default boot slot with configurable auto-boot timeout. Set a slot as default in the Edit/Re-order/Delete menu (F6), set the timeout in the Configuration menu (F4, Off/1/3/5/10 sec). If configured, the boot menu shows a countdown screen before the normal menu; any keypress cancels it, letting the timeout expire boots the default slot automatically.
- Menu slots can now hold just a disk mount and/or a BASIC command, with no program to launch — useful for "auto-mount a disk on power-on" style slots. Create one via the Filebrowser's **A/B** mount actions, or via **F2** (Edit command) on an empty slot in the edit menu.
- Adding a drive A image to a slot that already launches a program now warns first, since it replaces the disk the program expects to boot from.
- Fixed: auto-boot could hang in an endless loop when the default slot was at a letter position (A–H).
- Fixed: startup could fail to detect an installed REU (worked around a code generation issue in the current Oscar64 compiler).
- Fixed: the filebrowser's side status panel could show garbled/misplaced text under certain UCI mode and mount-state combinations (same underlying compiler issue).
- Various other stability and bug fixes.

Version 2.0.1 - 202604230

- Fix for freeze at startup (error in cartridge init code)

Version 2.0.0 - 20260429:

- Complete rebuild using the [Oscar64](#) cross-compiler (replaces cc65)
- Directory listings stored in REU memory, enabling:
 - much larger directory support
 - long filename support: maximum filename size now 50
 - long pathname support: maximum pathsize now 255. NB: The maximum is for the complete path, not for every directoryname in the path separately.
- Configurable colour scheme (F5 → F6 in configuration menu)
- Verbose or silent startup option (F5 → F2 in configuration menu)
- Splash screen in Information menu
- Configuration upgrade tool (uboot_upd12.prg) included for users upgrading from v1
- Various stability and bug fixes

Version 0.91 - 20230922:

- First public alpha (v1, cc65 build)

Building from source

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Build tooling

Tool	Purpose	Install
Oscar64	C cross-compiler targeting the 6502	Build from source; see the Oscar64 repo for instructions
zip	Bundles the release ZIP	<code>sudo apt install zip</code>
wput	FTP upload for make deploy	<code>sudo apt install wput</code>
curl	Reachability check before deploy	<code>sudo apt install curl</code>
pandoc	Regenerates README.pdf from README.md (optional — make all warns and skips if absent)	<code>sudo apt install pandoc</code> <code>texlive-xetex</code>

Getting the source

The Ultimate Command Interface library is a git submodule ([ultimate-uci-oscar64](#), in lib/ultimate-uci-oscar64), so clone with `--recursive`:

```
git clone --recursive https://github.com/xahmol/UBoot64-v2.git
```

or run `git submodule update --init` after a plain clone.

Deployment configuration

`make deploy` uploads the built cartridge and upgrade tools straight to your Ultimate devices over FTP, into `<storage>/Dev/build/`. The device IPs are kept out of git in a local `.env` file. `ULTIP2`

is optional; the storage defaults to usb0:

```
# .env (gitignored, never committed)
ULTIP1 = 192.168.1.xx
ULTUSB1 = sd
ULTIP2 = 192.168.1.yy
```

`make e2e` runs the end-to-end test on the same devices (or on `E2E_DEVICES`, if set); see [tests/e2e/README.md](#).

Make targets

Target	Effect
<code>make all</code>	Builds <code>uboot64.crt</code> , <code>uboot_upd12.prg</code> , regenerates <code>README.pdf</code> , and bundles the release ZIP — all into <code>build/</code>
<code>make clean</code>	Removes everything in <code>build/</code>
<code>make docs</code>	Regenerates <code>README.pdf</code> only
<code>make check-deploy</code>	Pings the configured Ultimate device without deploying
<code>make deploy</code>	Rebuilds if needed, then uploads <code>uboot64.crt</code> and both upgrade tools via FTP to every configured device (requires <code>.env</code>)
<code>make e2e</code>	End-to-end test on real hardware against golden screen dumps; backs up and restores your config (<code>make e2e-update</code> writes new goldens, <code>make e2e-restore</code> repairs an interrupted run)

Instructions

Prerequisites

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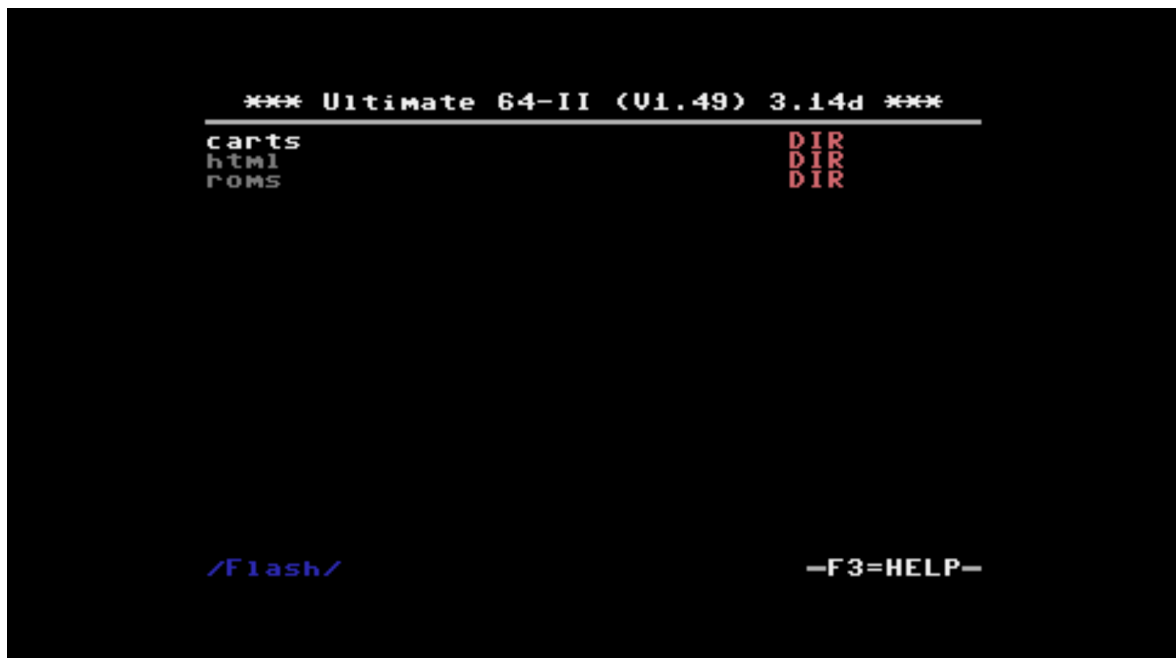
- UltimateII+ (U2+) cartridge installed on a real C64, or an Ultimate 64 / Commodore 64 Ultimate
- For Gideon's Logic products: Firmware at version 3.4 or higher (to have access to the UCI `DRVINFO` command — [firmware page](#))
- For Commodore Ultimate products: Supported starting from initial firmware
- RAM Expansion Unit (REU), minimum 128 KB, needs to be enabled
- Ultimate Command Interface needs to be enabled
- USB storage (up to 3 drives) or an SD card, for the configuration files

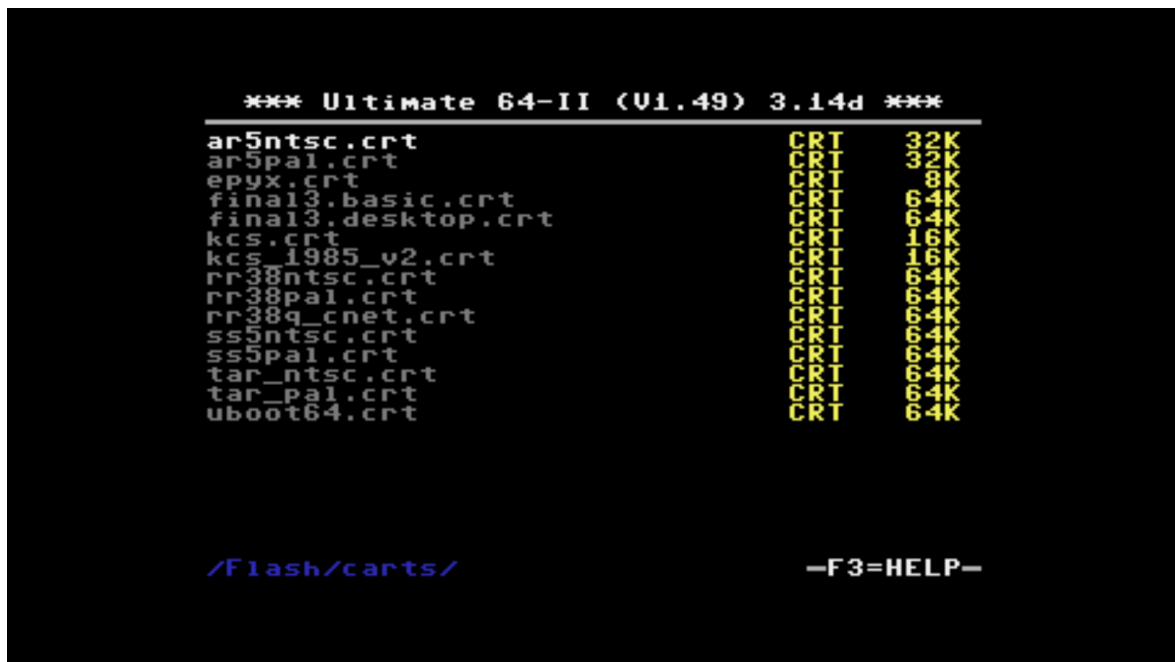
Installation

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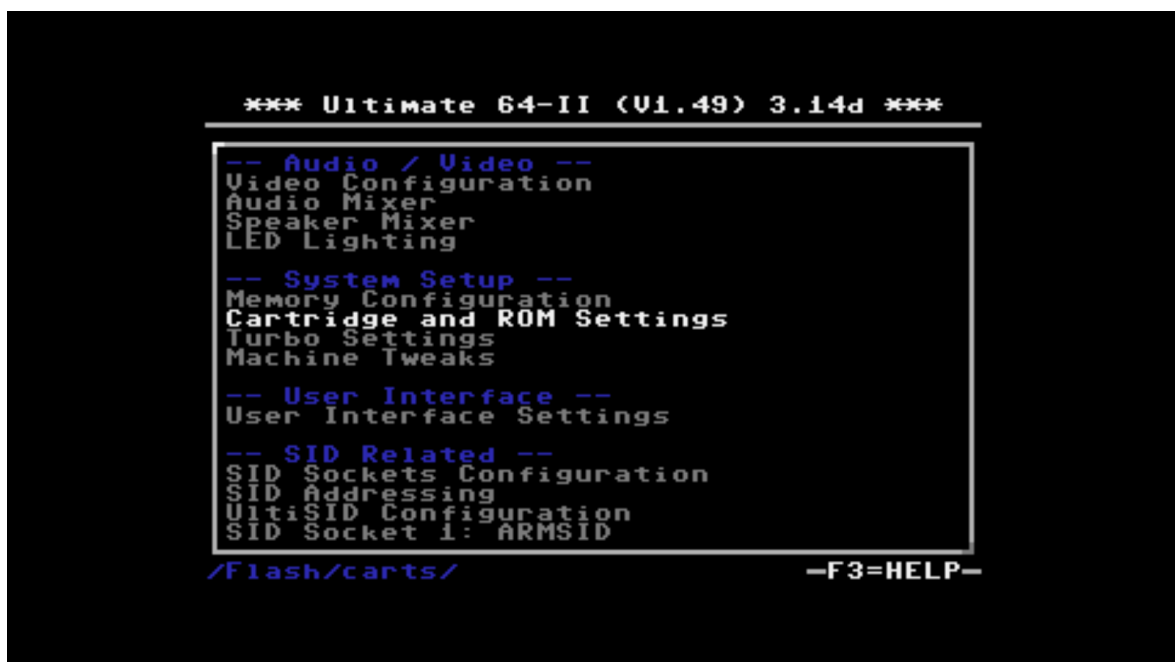
- Download the ZIP file with the latest build

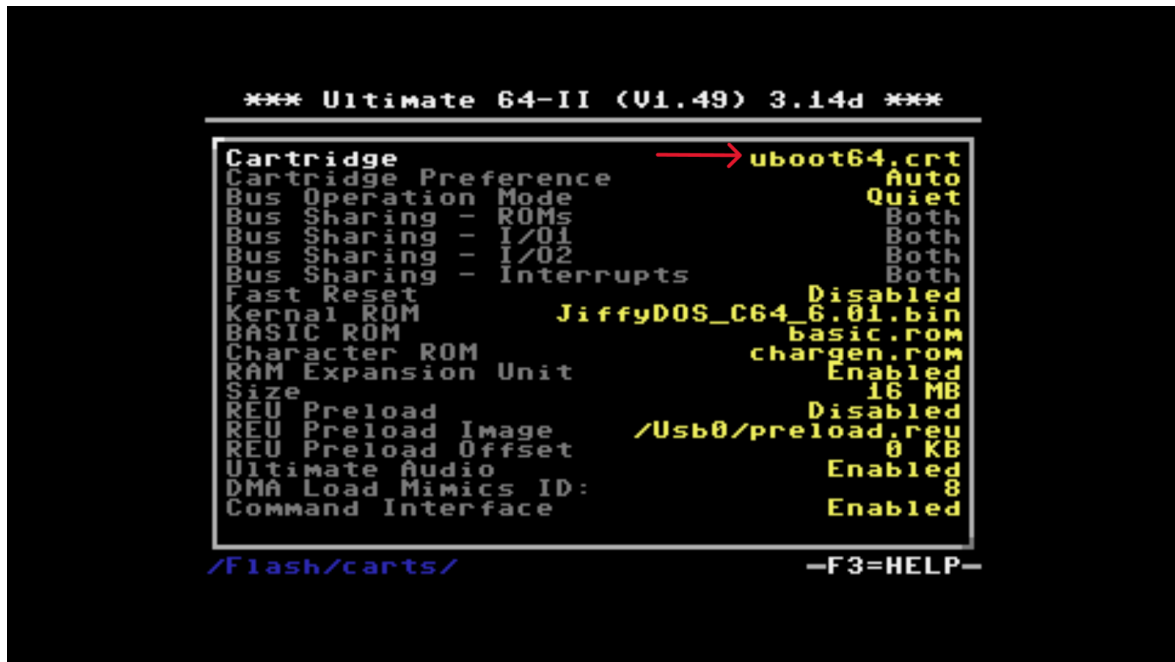
- In the ZIP file you will find these instructions, a .crt file (the cartridge image) and a uboot_upd12.prg file (configuration upgrade tool for v1 users). Unzip contents to a temporary location.
- Transfer the .crt file to the /Flash/Carts directory on your Ultimate device. Either transfer via FTP, or place it on the USB storage, browse to it in the UI filebrowser, press C= + C to copy, navigate to /Flash/Carts and paste with C= + V.



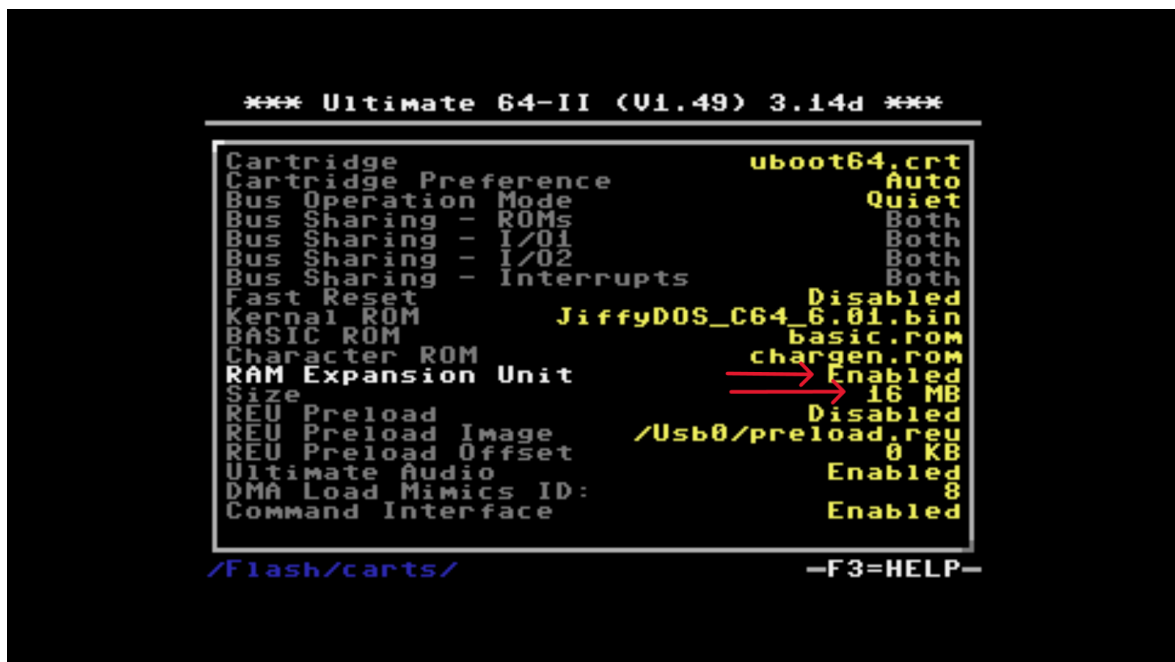


- Select the .crt file as the cartridge to start automatically: in the Ultimate UI press **F2**, navigate to **Cartridge and ROM Settings**, press **RETURN** twice to edit the **Cartridge** option, and select **uboot64.crt** from the drop-down.

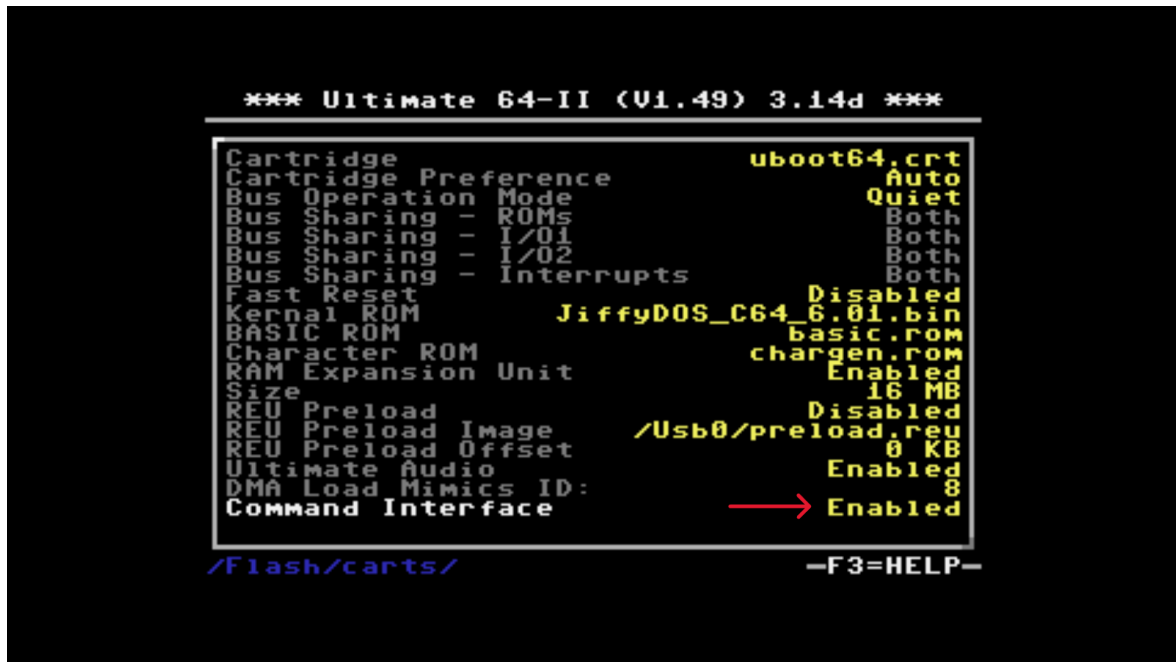




- In the same **Cartridge and ROM Settings** menu, ensure that RAM Expansion Memory (REU) is enabled. Size does not matter as even the smallest possible size of 128KB will be sufficient.



- In the same **Cartridge and ROM Settings** menu, ensure that the Ultimate Command Interface is enabled.

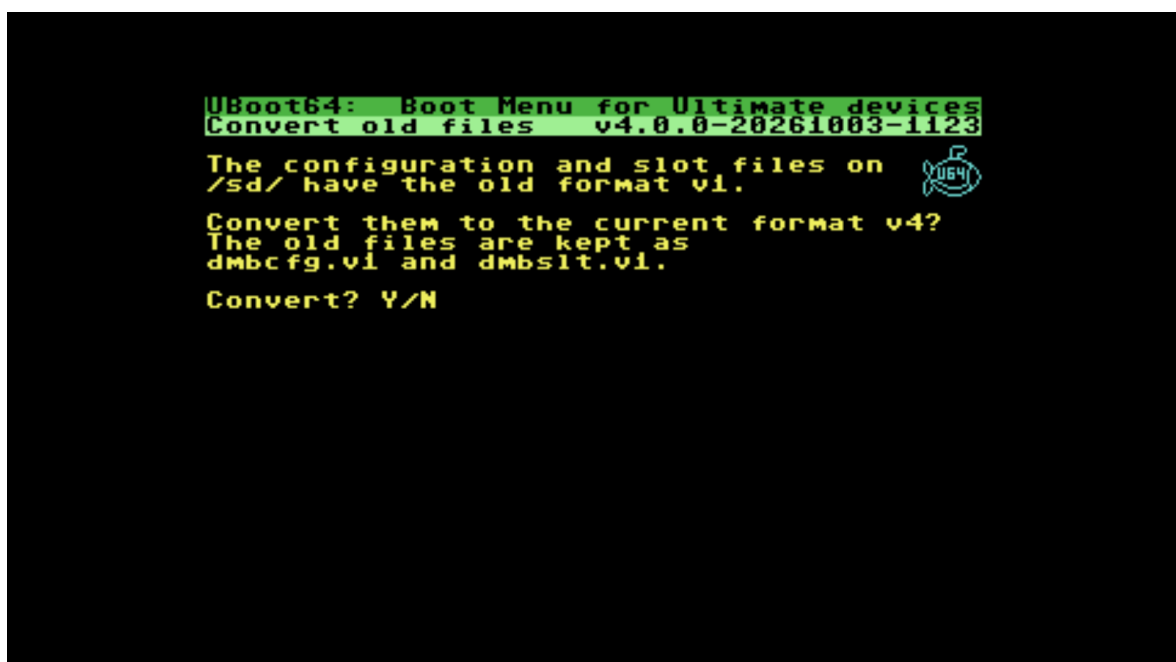


- Press **STOP** to leave the configuration menu and **RETURN** to confirm saving to flash.
- Power cycle your device. UBoot64 should now start automatically.
- Stop autostarting by deselecting the cartridge via the same procedure and choosing **None** or another image.

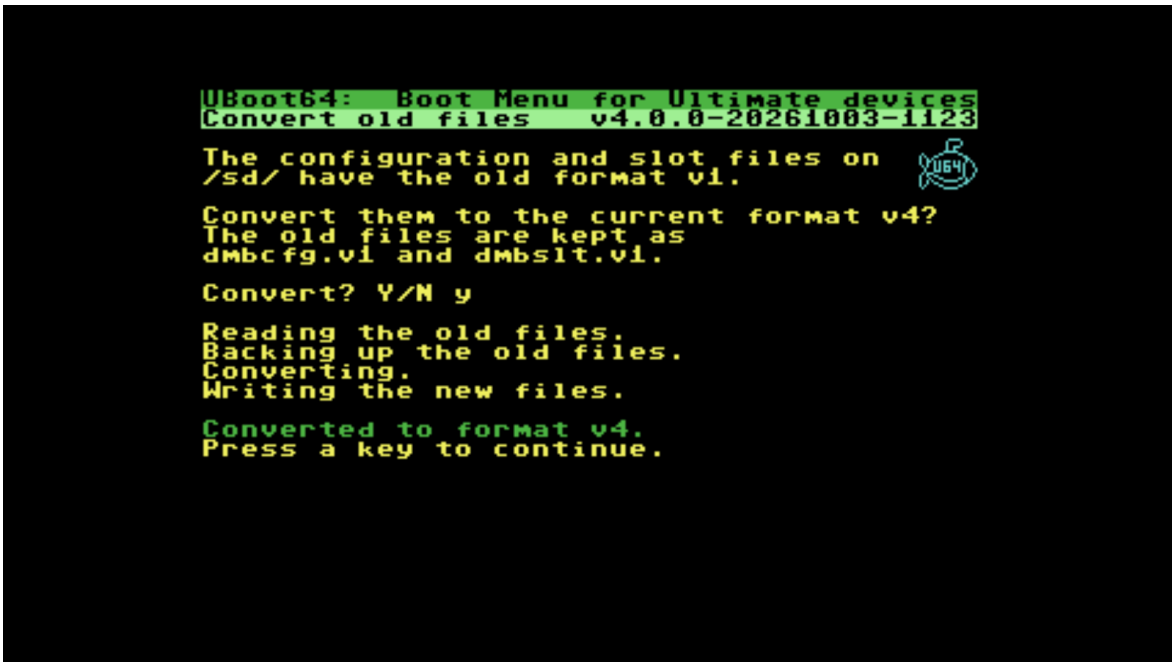
Upgrading from an earlier version

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From v4.0.0 UBoot64 converts the configuration and slot files of earlier versions itself: format v1 (UBoot64 v1), v2 (v2.x) and v3 (v3.0.x). v4.0.0 uses format v4, which adds a settings file per slot. When UBoot64 finds files in an older format at start-up, it asks:



- **Y** first copies the old files to `dmbscfg.v1/dmbslt.v1` (or `.v2`, `.v3`) next to them and checks the copies, then writes the files in the new format and continues the start. Your slots and settings are kept (from v1: the slots, NTP server and time offset; v1 had no other settings). REU preload slots from v1 and v2 get their REU image path in its own field.
- **N** changes nothing and exits to BASIC; you are asked again at the next start.
- If a step fails, UBoot64 says which one. Before the new files are written your files are unchanged; after that, the copies hold the old files.
- A REU of at least 128 KB is needed for the conversion (UBoot64 needs a REU anyway).



```

UBoot64: Boot Menu for Ultimate devices
Convert old files v4.0.0-20261003-1123

The configuration and slot files on
/sd/ have the old format v1.

Convert them to the current format v4?
The old files are kept as
dmbscfg.v1 and dmbslt.v1.
Convert? Y/N y

Reading the old files.
Backing up the old files.
Converting.
Writing the new files.

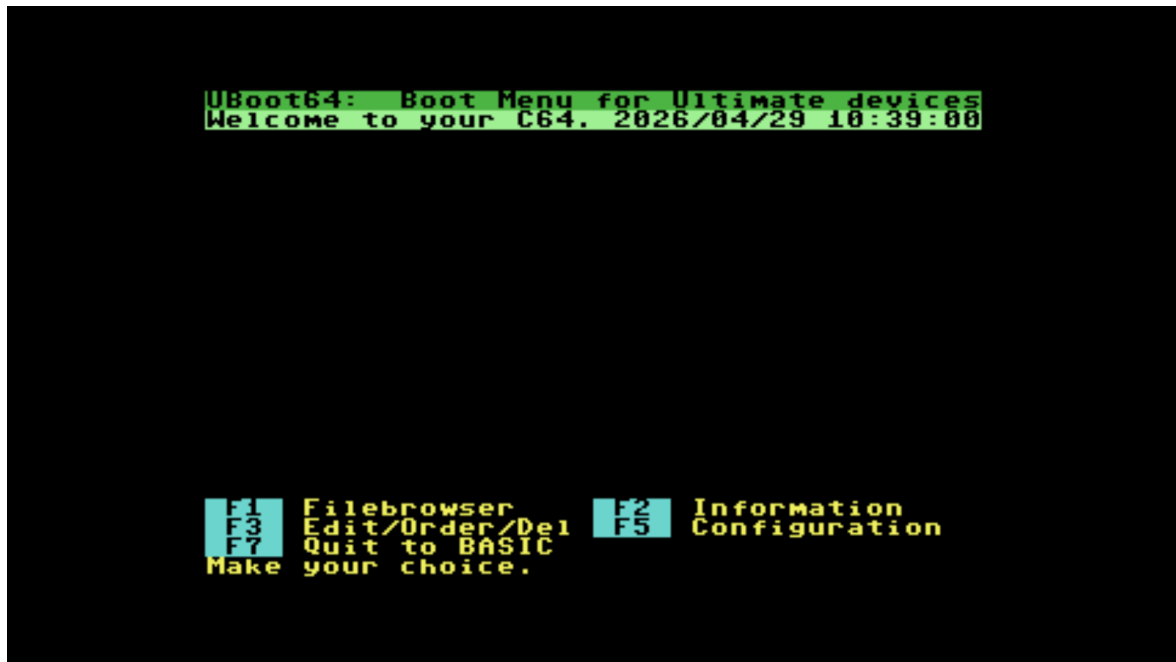
Converted to format v4.
Press a key to continue.
  
```

The separate upgrade tools still work, for this release: `uboot_upd12.prg` (v1) and `uboot_upd23.prg` (v2) convert to the current format too, without keeping copies. Run them from the Ultimate UI or a BASIC prompt before starting UBoot64. There is no tool for v3: the cartridge does it.

First run

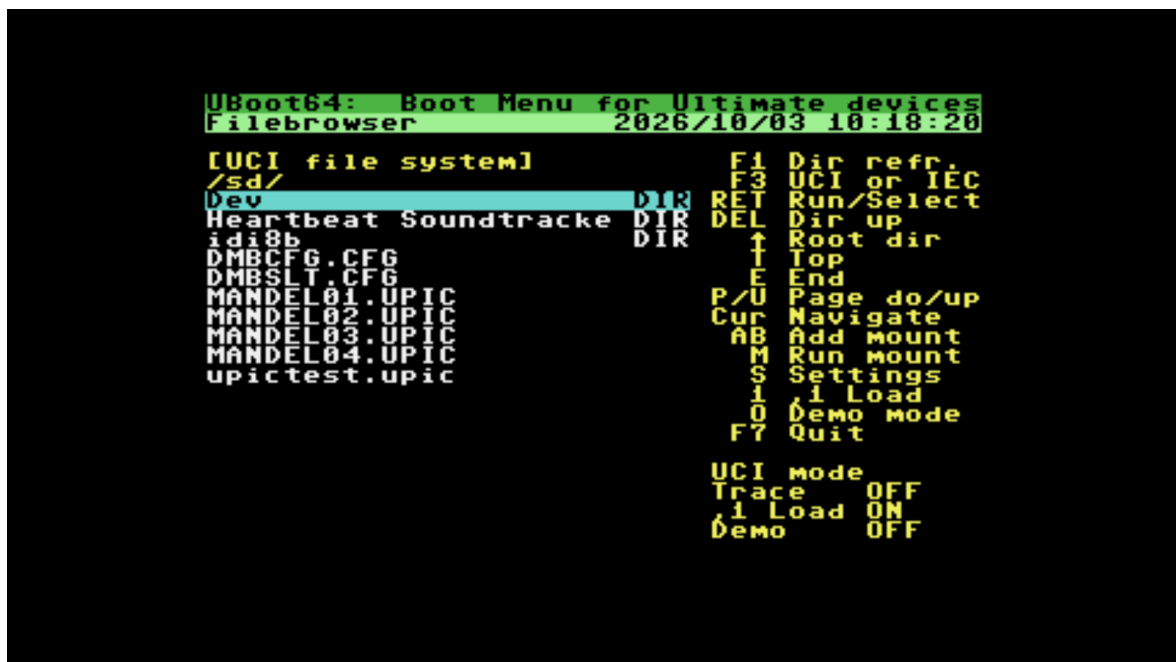
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- At first run, no configuration file is present yet, so UBoot64 creates configuration files with an empty start menu and default settings.
- UBoot64 looks for existing configuration files on the SD card first, then on the USB storage devices (every device the Ultimate lists, such as USB0, USB1). The first one where they're found is used; if none have them, they're created on the first of those that's actually present (same order). The files are `DMBCFG.CFG` (configuration data) and `DMBSLT.CFG` (menu slot data), always at the root of that device.
- On startup a splash screen is shown briefly. Press any key to proceed to the main menu.
- The empty start menu looks like this:



Add start options via the Filebrowser

- Start options can be added to menu slots **0–9** and **A–Z** (18 slots total) via the Filebrowser.
- Press **F1** to open the filebrowser:



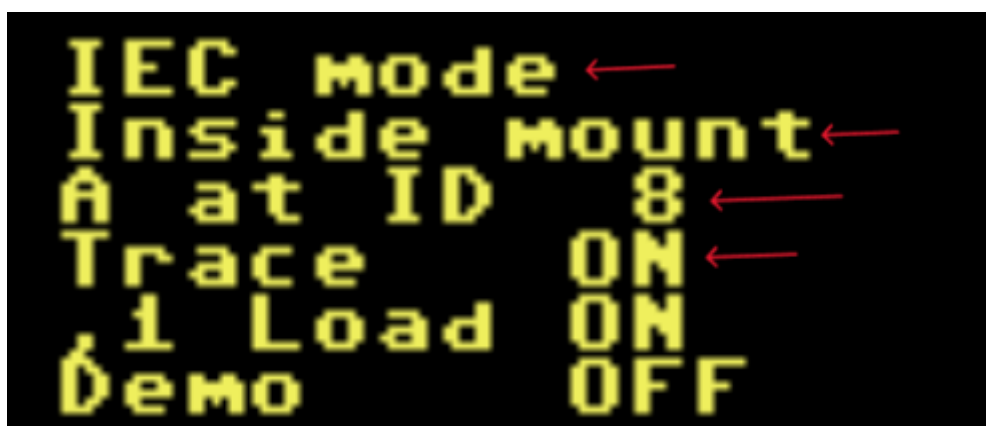
- The browser starts in **UCI mode** — browsing via the Ultimate Command Interface on the native filesystem of your Ultimate device.
- Press **F3** to switch to **IEC mode** for drives connected to the IEC bus. Use + / - to select the device number.



- Press **F1** to refresh the directory if needed.
- In IEC mode, press **D** to start a **directory trace** from the root, needed to record the full path for a menu slot. The TRACE toggle in the lower right switches to ON.



- **,1 Load** (toggle with 1): loads with LOAD"NAME", 8 (off) or LOAD"NAME", 8, 1 (on).
- **Demo mode** (toggle with O): powers down all drives except ID 8 on start.
- Navigate with **cursor keys** (UP/DOWN in dir, ENTER/RIGHT to enter dir or image, DEL/LEFT to go up).
- From UCI mode, entering a disk image (.Dxx extension) mounts it on drive A and switches to IEC mode — "Inside mount" appears in the menu.



- Press **ENTER** on a program file to select it for a menu slot. In IEC mode inside a disk image mounted from UCI mode, the associated drive A disk image is automatically included.
- In UCI mode, press **ENTER** on a loose .prg file to add it to a slot as it is: no drive or disk image needed (see [Programs from UCI mode](#) below).
- Press **A** or **B** on a disk image (.Dxx) to select it as the additional image to mount on drive A resp. B when starting from that slot.
- Press **M** on a program file to select it to be run from the disk image already mounted on drive A.
- To add a REU file: navigate to it and press **ENTER**, then select the target slot, choose REU size with + / -, and confirm with **ENTER**.



Note: The REU file must be in the same filepath as the drive A disk image if one is present.

- After selecting what to place in a slot, you are shown this screen to pick the slot position:



- If the slot is already filled, confirmation is requested. Choose **Y** to proceed or **N** to cancel.
- Press **0–9** or **A–Z** to choose the slot, enter a name and press **ENTER**.
- Repeat until all desired slots are filled.
- Important notes:
 - If you want to add a program, disk images and a REU image to the same slot, this needs to be done in separate steps for each file or image to add. Just choose the same slot again each time.

Programs from UCI mode From v4.0.0 a loose .prg file in UCI mode can be added to a slot directly: highlight it, press **ENTER** and pick the slot. At boot UBoot64 loads the file through the Ultimate Command Interface into the REU, and copies it into C64 memory just before handing over to BASIC, then types RUN. No drive, disk image or IEC bus is involved, so this works on every Ultimate.

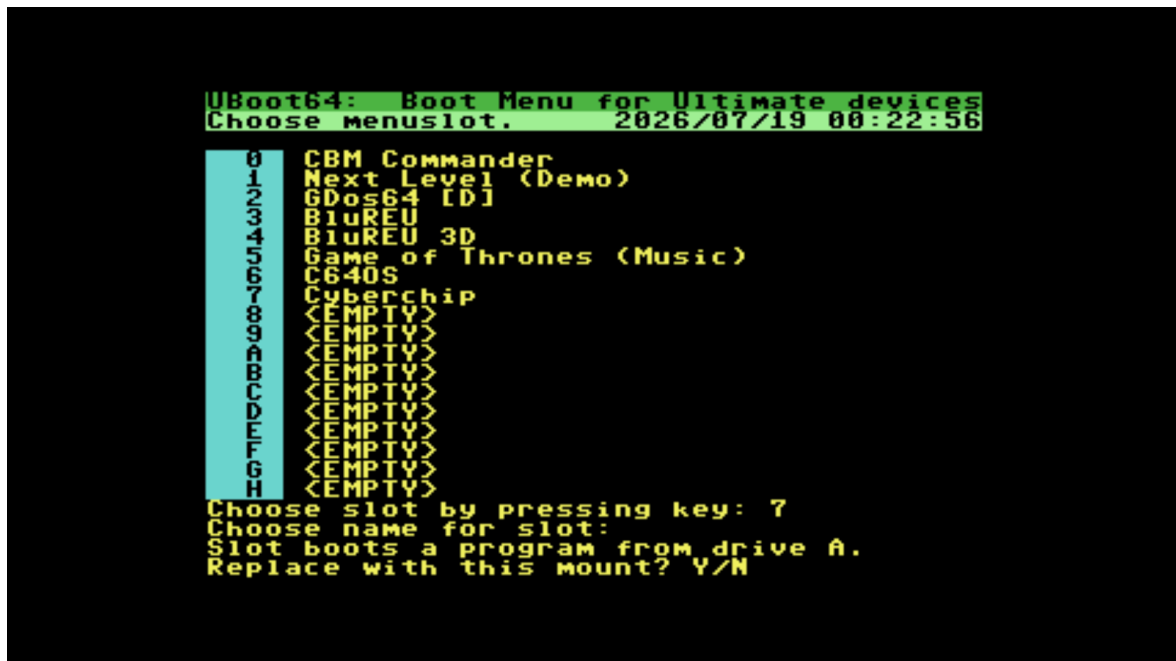
- With **,1 Load** off the program goes to \$0801, as LOAD"NAME", 8 would; with it on, to the address in the file, as LOAD"NAME", 8, 1.
- A BASIC command of the slot (**F2** in the edit menu) is typed before RUN.
- A settings file <program>.cfg next to the program is applied automatically (see [Program settings files](#)).
- Limits: the program must not overlap C'000–C'0FF, where UBoot64's last bit of code runs while it copies the program. A REU of at least 128 KB is needed (the program is staged in the last 64 KB); a REU preload image that fills the whole REU can't be combined with such a program in one slot.

Mount/command-only slots A menu slot does not need a program to launch. Pressing **A** or **B** on a disk image without ever selecting a program to run creates a slot that only mounts that disk and then

lands at the BASIC READY. prompt on boot — the classic "auto-mount a disk on power-on" setup, especially combined with a [default boot slot and timeout](#).

You can also create a slot that runs only a BASIC command, with no disk mount and no program: in the edit menu (**F3**) press **F2** (Edit command) and pick an **empty** slot — you'll be prompted for a name, then the command.

These combine freely: adding a drive B image or a REU preload to a slot that already launches a program keeps the program intact (useful for programs that expect a second disk or preloaded data). Adding a drive A image to a slot that already launches a program **replaces** the disk the program expects to boot from, so you'll be asked to confirm first — if you just want an extra mount alongside an existing program, add it as drive B instead, or use a fresh slot for a drive-A-only mount.



F1: Filebrowse menu

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The filebrowser is based on and inspired by the DraBrowse program from <https://github.com/doj/dra-copy>



Key reference

Key	Function
F1	Read / refresh directory
F3	Toggle between UCI and IEC modes
F4	Show the partition list (IEC mode only, on a device that supports partitions)
+	Increase device number (IEC mode only)
-	Decrease device number (IEC mode only)
RETURN	Enter directory / run selected program (IEC mode) / select for slot; on a .prg in UCI mode: add it to a slot as a program from UCI mode
CURSOR RIGHT	Enter directory
DEL	Go to parent directory
CURSOR LEFT	Go to parent directory
↑	Go to root directory
T	Go to the first item in the directory
E	Go to the last item in the directory
P	Page down in the directory
U	Page up in the directory
Cursor keys	Navigate in the directory
D	Toggle Dirtrace (IEC mode only): traces directory movements from root
A	Select highlighted image (.Dxx) to be mounted on drive A

Key	Function
B	Select highlighted image (.Dxx) to be mounted on drive B
M	Select highlighted program to run from the disk image on drive A
S	Select the highlighted settings file (.cfg or .usr) as a slot's own settings file, applied when the slot boots (see Program settings files). UCI mode, and IEC mode on the Ultimate's SoftIEC drive (firmware 3.15+)
1	Toggle ,1 load
O	Toggle demo mode
Q or F7	Quit to main menu

UCI mode and IEC mode The file browser opens in **UCI mode** by default. Press **F3** to toggle between modes. The current mode is shown in the lower right of the sidebar.

UCI mode browses the native filesystem of the Ultimate device (USB storage) directly via the Ultimate Command Interface. It supports long filenames, full directory paths, and can enter disk images (.D64, .G64, .D71, .G71, .DNP) as if they were directories.

IEC mode reads directories from a physical or emulated IEC bus device (disk drives on the serial bus). Press + or - to cycle through available active device numbers. In IEC mode, filenames follow CBM conventions and directory listings are read via the KERNAL. File type labels (PRG, SEQ, DEL, REL, CBM, DIR) are shown next to each entry.

Directory navigation Use **CURSOR UP** / **CURSOR DOWN** to move the highlight one entry at a time. Use **P** (page down) and **U** (page up) to jump a full screen at a time. **T** jumps to the first entry in the directory and **E** jumps to the last.

To enter a subdirectory or disk image, press **RETURN** or **CURSOR RIGHT** on it. To go back up to the parent directory, press **DEL** or **CURSOR LEFT**. Press ↑ (the up-arrow key, not cursor up) to jump directly to the root of the current file system.

Press **F1** at any time to re-read and refresh the current directory listing.

Directory trace (IEC mode) Because IEC drives do not report their full path, pressing **D** activates **directory trace** mode: UBoot64 records each directory you navigate into, building the full path from root. The TRACE indicator in the sidebar switches to **ON**. Navigate from the root directory with trace active before selecting a program, so the complete path can be stored in the menu slot.

Without trace active in IEC mode, only the filename is stored; the path will be empty and the program can only be found if it is in the root directory of the drive.

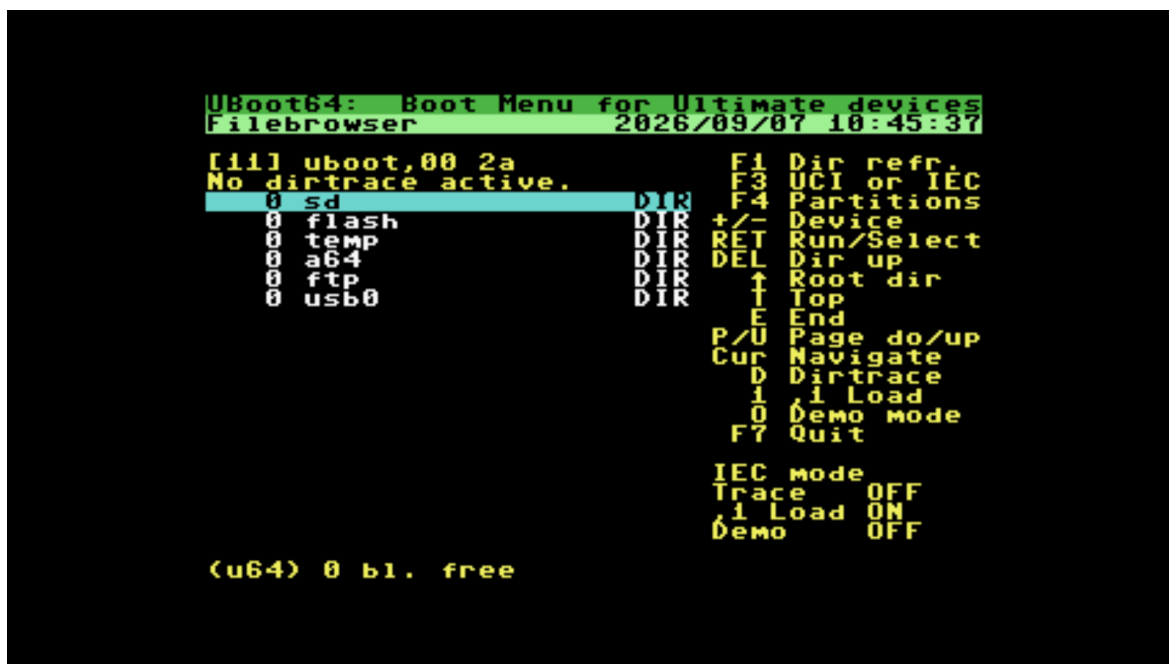


Press **D** again to turn trace off and reset the recorded path depth.

Partition browsing (IEC mode) Some IEC devices — the Ultimate's own SoftIEC on firmware 3.15+, and other partition-capable devices such as CMD-HD or SD2IEC — support multiple partitions on a single device number. Press **F4** to show a list of the device's partitions instead of the normal directory. Select one with **RETURN** to browse into it.



While inside a partition, pressing **DEL** at that partition's own root shows the partition list again, rather than trying to leave the device entirely — press **DEL** again from a subfolder to go up one level first, as normal.



If the current device doesn't support partitions, **F4** shows a "Could not read partition list" message instead.

Toggles: ,1 Load and Demo mode The sidebar shows two additional toggles that affect how a program is launched when selected for a slot.

,1 Load (toggle with **1**): controls the BASIC LOAD command used at boot time.

- **OFF**: LOAD"FILENAME", 8 — loads the program to the address in the file header (normal for most games and programs).
- **ON**: LOAD"FILENAME", 8, 1 — loads to the absolute address stored in the file. Needed for some utilities and programs that require loading at a fixed memory address.

Demo mode (toggle with **O**): when ON, all Ultimate emulated drives except the one on IEC ID 8 are powered off before the program starts. Many demos and some games require only one drive to be active. Enable this toggle when adding demos to ensure reliable playback on systems with multiple emulated drives.



Both toggle states are stored in the menu slot when you confirm the selection.

Selecting items for menu slots After navigating to the desired file or image, use the appropriate key for the type of item:

Selecting a program to run (RETURN or M):

- In IEC mode, pressing **RETURN** on a .PRG file without trace active runs it immediately (direct launch, no slot storage).
- In IEC mode with trace active, pressing **RETURN** on a .PRG stores it in a slot — the recorded path and filename are saved.
- In UCI mode, pressing **RETURN** on a .PRG file does nothing unless first a disk image was entered. If a disk image was entered before (since mount), the program stores it in a selected slot and also automatically stores that disk image as the drive A image.
- Pressing **M** instead of **RETURN** stores the program to run from the disk image already mounted on drive A (IEC inside-mount mode). Use this when the program must be started from within the mounted disk.

Entering and using a disk image (RETURN on .Dxx file in UCI mode): Pressing **RETURN** on a disk image file in UCI mode mounts it on drive A and switches to IEC mode, showing "Inside mount" in the sidebar. You can then browse the disk contents and press **RETURN** on a program to store both the program and the disk image in a slot.

Adding an additional drive B image (B): In UCI mode, pressing **B** on a disk image (.Dxx) stores it as the drive B image for the next slot you choose. This is useful when a program needs two drives simultaneously.

Adding a drive A image without entering it (A): Pressing **A** on a disk image in UCI mode stores it as the drive A image for the next slot, without mounting or entering the image. Use this when you want to add an image to a slot separately from the program selection step.

Selecting a REU image (RETURN on .REU file): Navigate to a .REU file and press **RETURN**. You are then prompted to choose the REU size with + / - before confirming. The REU image is preloaded into REU memory before the program in the slot starts.

After any selection, you are taken to the slot picker screen to choose which of the 18 menu slots (0–9, A–H) the item should be stored in.

Limitations in UCI mode Loose .PRG files on the native UCI file system cannot be added to a menu slot for direct launch. The Ultimate Command Interface does not (yet) support launching a program directly from a USB path. Options:

- Use IEC mode via SoftIEC to access the file through a CBM device number.
- Enter a disk image in UCI mode first, then select the program from within the mounted image.
- Use the Ultimate's own UI for running programs directly from USB.

F2: Information

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Shows first the splash screen, and after pressing any key, the information and credits screen, with the UBoot64 version and the Ultimate it runs on (the product name the firmware reports, or "unknown"). Press any key to return to the main menu.



F3: Edit / re-order / delete

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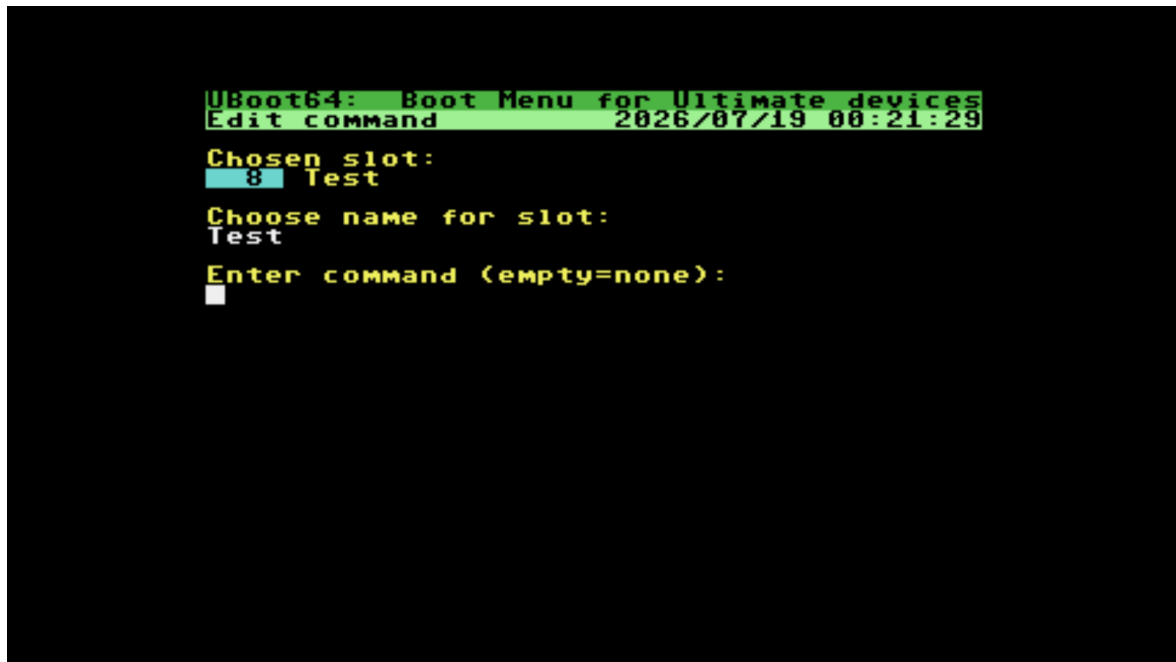
Rename, re-order, edit commands for, or delete menu slots.



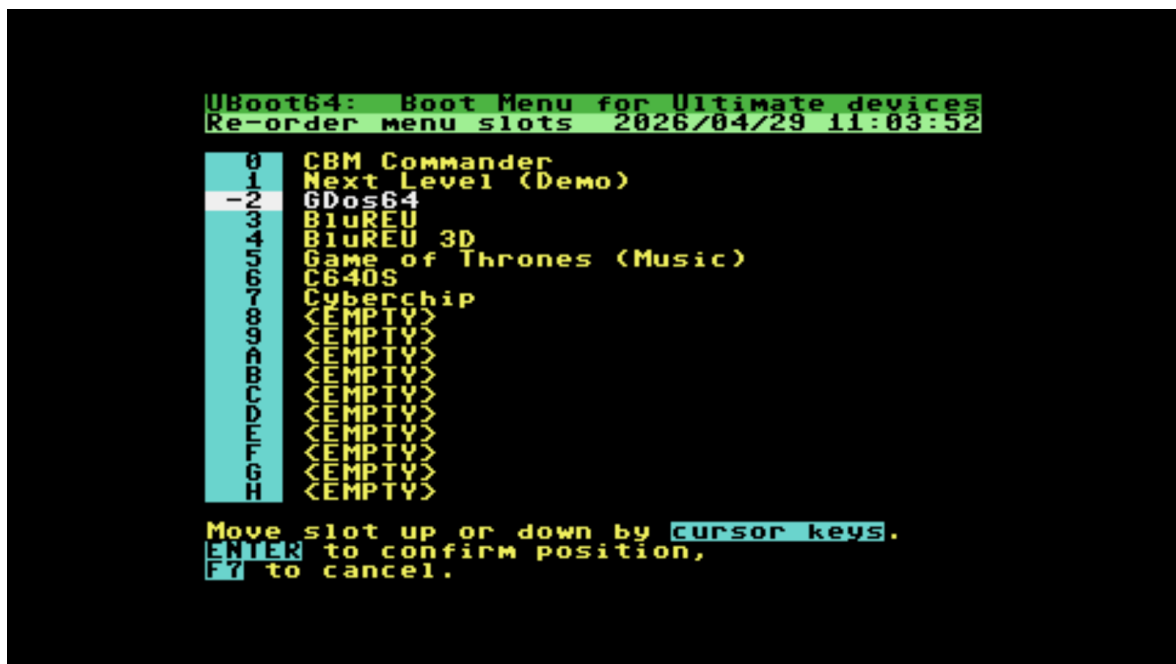
- **F1** — Rename a slot. Choose slot (**0–9 / A–Z**), enter new name, press **ENTER**.



- **F2** — Add or edit a user-defined BASIC command to execute before the program starts.



- **F3** — Re-order slots. Choose slot to move (**0–9 / A–Z**), highlighted in white. Press **UP / DOWN** to move. Confirm with **ENTER**, cancel with **F7**.



- **F5** — Delete a slot. Choose slot (**0–9 / A–Z**), confirm with **Y** or cancel with **N**.

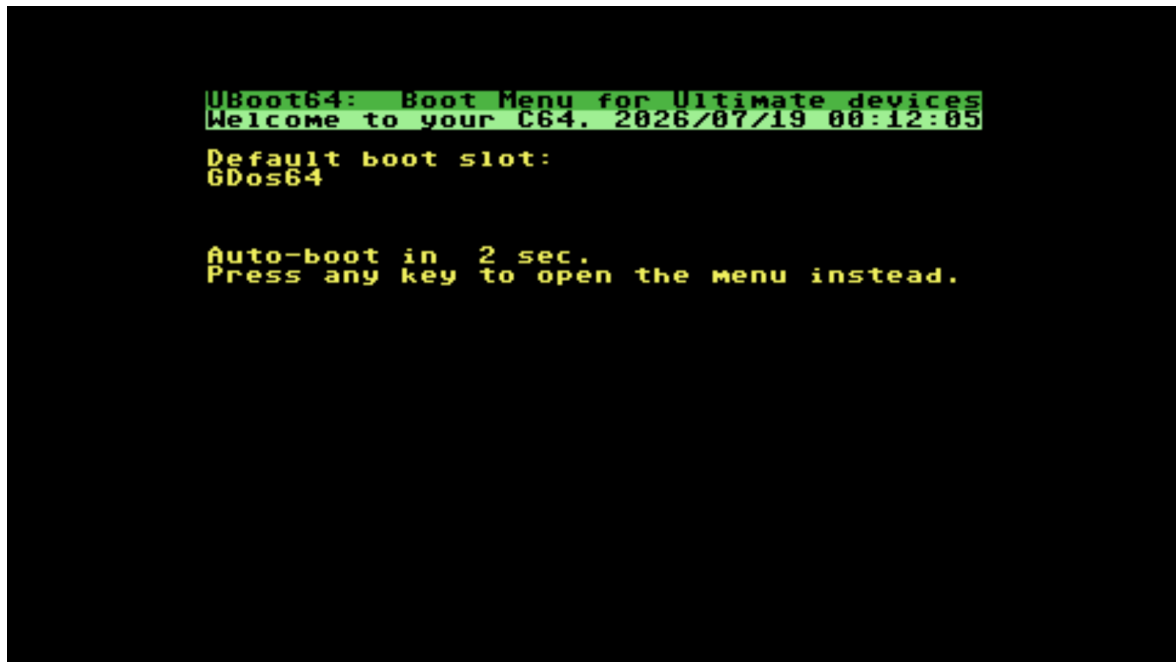


- **F6** — Set or clear the default boot slot. Choose a slot; slots currently marked default are shown with a [D] tag in the list. Choosing the slot that already has [D] clears it; choosing another slot moves the default there (only one slot can be default at a time). See [Default boot slot and auto-boot timeout](#).
- **F7** — Return to main menu. Changes are saved at this point.

Default boot slot and auto-boot timeout

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You can mark one menu slot as the default boot target and have it launch automatically after a count-down, without needing to press a key — useful for auto-mounting a disk (or launching a program) on power-on.



- Set the default slot: main menu → **F3** → **F6**, then pick a slot. Slots currently marked default show a [D] tag wherever slots are listed.



- Set the timeout: main menu → **F5** → **F4** to cycle Off / 1 / 3 / 5 / 10 seconds. Default is **Off**, so this has no effect until both a default slot and a timeout are configured.
- When both are set, the boot menu shows a dedicated countdown screen (rather than the full menu) with the default slot's name and remaining seconds. Pressing **any key** cancels the countdown and opens the normal menu — the keypress itself is not treated as a selection. If the countdown reaches zero, the default slot boots automatically, exactly as if you had pressed its key.
- The default slot follows the slot's content through renames and re-ordering. Deleting the slot (or picking it again via **F6**) clears the default marker.

- A default slot does not need to launch a program — see [mount/command-only slots](#) below for the classic "auto-mount a D64 on power-on" use case.

USB port auto-reroute

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A menu slot's disk image (drive A / B) and REU file are stored with the USB port they were on at the time (e.g. /usb0/ . . .). If you later plug that stick into a different USB port, the slot would previously fail to boot.

UBoot64 now detects this automatically at boot time: if the stored path and file aren't found there, it retries on the other USB ports (USB0/1/2). If found, it shows a brief "Rerouted to ..." message and continues booting from there — this only affects the current boot and is not saved back to the slot, so the slot still uses its originally configured port next time (letting you use multiple sticks across different slots as before). This checks the actual file, not just the folder — if two sticks happen to share a folder name (e.g. both have a games folder) but only one has the specific file the slot needs, UBoot64 correctly finds and boots from the right one instead of stopping on the wrong stick.

If the file isn't found on any USB port, boot stops with a message asking you to insert the stick — press any key to retry once it's plugged back in, or **F7** to give up and return to BASIC (useful if you meant to boot something else entirely).

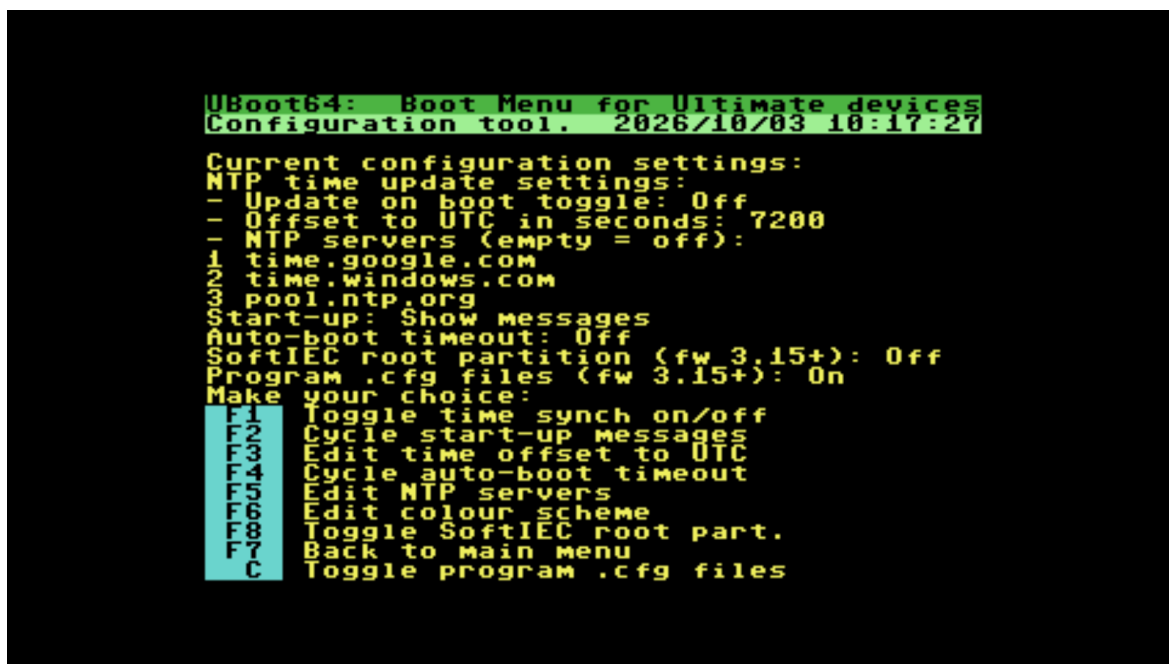
This only applies to disk image and REU paths (which use USB port addressing via the Ultimate Command Interface). It does not apply to programs launched directly from an IEC device without a disk mount, or to the SD card slot, since neither uses a swappable USB port.

F5: Configuration

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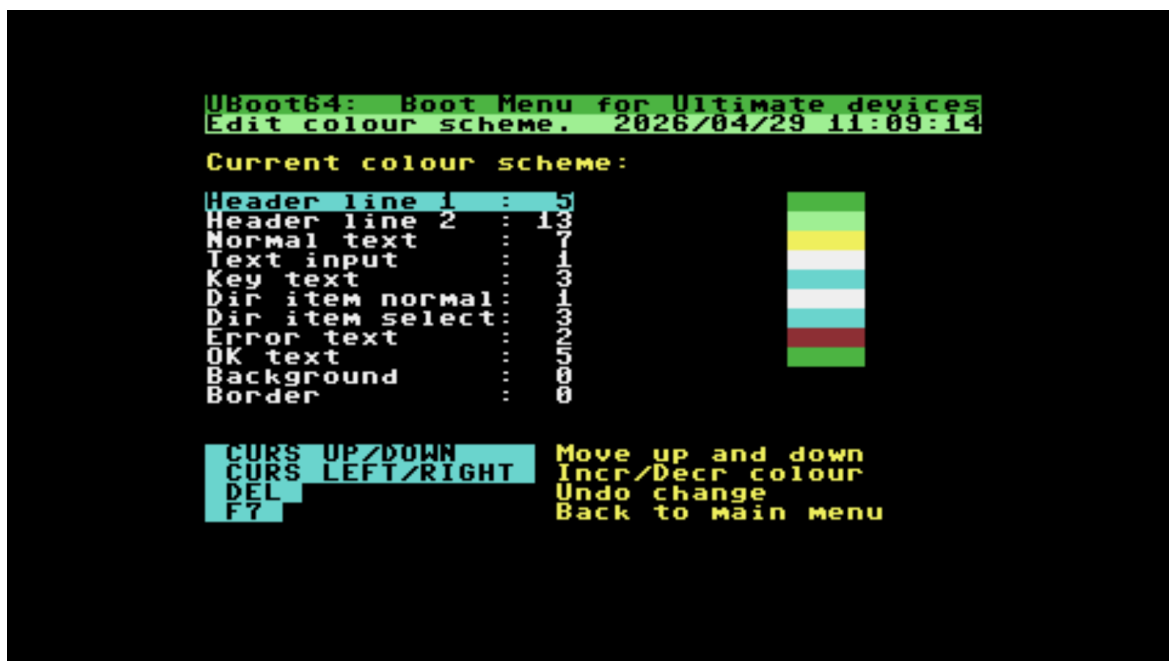
Configure NTP time synchronisation, startup verbosity, and the UI colour scheme.

After pressing **F5**:



The screen shows current settings and allows editing:

- **F1** — Toggle NTP time sync on or off. Default: off, since the firmware of the Ultimate 64 Elite II and the C64 Ultimate (3.14d and later) sets the clock itself. Switch it on for older firmware or devices without their own time sync.
- **F2** — Cycle the start-up messages: **Silent** (only a progress indicator), **Show messages** (detailed feedback during startup), or **Show messages + wait** (the same, then waits for a key so you can read them). Default: Show messages.
- **F3** — Edit the time offset to UTC in seconds. Automated daylight saving adjustment is not provided; adjust manually when needed. Examples: CET = 3600, CEST = 7200. See <https://www.timeanddate.com/time/zones/> for all offsets (multiply hours by 3600). Default: 7200 (CEST).
- **F4** — Cycle the auto-boot timeout: Off, 1, 3, 5, or 10 seconds. Default: Off. See [Default boot slot and auto-boot timeout](#).
- **F5** — Edit the three NTP servers, one after the other. They are tried in order until one answers; an empty server is skipped. **RUN/STOP** keeps a server as it was. Defaults: `time.google.com`, `time.windows.com`, `pool.ntp.org`. An existing configuration keeps its server as server 1 and gets the last two defaults as servers 2 and 3.
- **F6** — Edit the UI colour scheme. Use **CURSOR UP** / **DOWN** to select a colour element, **CURSOR LEFT** / **RIGHT** to change its colour value (0–15). **DEL** reverts to the saved colours. **F7** returns to the configuration menu. Changes are saved when you exit the configuration menu.



- **F8** — Toggle the "SoftIEC root partition" option (firmware 3.15+).

This option by itself does nothing. Pressing F8 only saves a preference; it does not touch the device. The actual effect only happens later, in the file browser, **the next time you press F3 to switch into IEC mode on the Ultimate's own SoftIEC drive** — that specific transition is the only place this option does anything. If you're already in IEC mode when you turn the option on, or you toggle it while browsing a different (non-SoftIEC) IEC device, nothing happens until you actually switch into IEC mode on the SoftIEC drive.

When that transition happens with the option **on**: UBoot64 auto-creates and selects a partition exposing the whole filesystem at root (/), useful if you haven't set up a partition yourself via the Ultimate's own menu. It never touches a partition you've already configured at that same slot (254) — if one exists there with a different path, UBoot64 shows an error and leaves it alone instead of overwriting it.

Turning the option **off** just stops UBoot64 from creating/selecting that partition on future **F3** transitions — it does not remove an already-created partition from the device. Default: off.

This partition is temporary regardless. UBoot64 creates it in the Ultimate's live, in-memory partition table — there's no way for a cartridge to make that permanent, only the Ultimate's own on-screen UI can save the partition table to flash (its "Save Partitions" action, which writes `iec_partitions.ipr`). Without that one-time manual save, the partition is gone after a power cycle either way, so there's nothing to actively clean up: if you want it gone, a power cycle (without a manual save) removes it, same as it always would have.

- **C** — Toggle "Program .cfg files" (firmware 3.15+): apply a program's own Ultimate settings file when its slot boots. Default: on. See [Program settings files](#).
- **F7** — Return to main menu. Changes are saved.

Program settings files

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The Ultimate can save its settings to a .cfg file ("Save Settings" in its menu). The file only needs the lines that matter, so a short file can, for example, set a drive type or switch off turbo for one game:

```
[Drive A Settings]
Drive Type=1581
```

With firmware 3.15 or later the Ultimate applies <program>.cfg by itself when you start a program from its own file browser. A program started from a UBoot64 slot doesn't pass through that browser, so UBoot64 looks for the file itself when it boots a slot (unless **C** in **F5** has switched this off):

- a slot that mounts a disk image on drive A: <image name>.cfg next to the image, e.g. gdos64.cfg next to gdos64.d81;
- a slot on the Ultimate's SoftIEC drive (made with the SoftIEC root partition, see **F8**): <program>.cfg next to the program.

You can also give a slot its own settings file: in the file browser, highlight a .cfg or .usr file, press **S** and pick the slot. This works in UCI mode, and in IEC mode on the Ultimate's SoftIEC drive (firmware 3.15+; there a .usr file is listed without its extension, as a **USR** file). So you can make a slot in IEC mode and add its settings file without switching modes. That file is then used instead of the automatic lookup above, for any kind of slot, also one on a real drive. If it's missing at boot, UBoot64 says so and boots anyway.

If there is no .cfg, UBoot64 tries <name>.usr, as the Ultimate does. The file can't be in the root of a USB stick or SD card: the Ultimate's firmware (3.15a) can't load settings files from there, so an image directly in /usb0/ gets no settings. The settings are applied first, before the disk images are mounted, so they can change drive types. The boot screen shows "Settings from ." If the file has a line the Ultimate can't apply, UBoot64 shows the Ultimate's message for 3 seconds and boots anyway. No file, or older firmware without this function: nothing happens.

The settings are not saved: they last until the Ultimate is switched off, or until something changes them again, like the settings applied by the Ultimate's own browser. Slots on a real drive, or on drive emulation without an image, have no Ultimate path, so no settings file is looked for.

Baseline settings: to set things back after a program, put a settings file uboot64.cfg in a folder uboot64 on the device that holds UBoot64's configuration (e.g. /usb0/uboot64/uboot64.cfg). UBoot64 applies it at every start, before the menu.

F7: Quit to BASIC

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Exit the boot menu to the BASIC Ready prompt. Memory is erased on exit.

Known issues

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- **No Ultimate Audio for programs started from UBoot64.** A program that uses Ultimate Audio (*DF20–DFFF*) doesn't find it when started from a UBoot64 slot, even with "Map Ultimate Audio *\$DF20–DFFF*" enabled in the Ultimate's settings (or in a program settings file). The Ultimate switches Ultimate Audio off as long as a cartridge of UBoot64's type (Final Cartridge III) is plugged in, and that holds until the C64 is reset without it. Start such programs from the Ultimate's own file browser instead. Under investigation: [GitHub issue #25](#).
- **Rare hang at start-up.** Very occasionally the start stops after the drive list (before "IDs needing manual power switching"), seen in automated tests right after old configuration files were converted. Restarting UBoot64 gets past it. [GitHub issue #24](#).
- **Settings files in the root of a storage device** can't be loaded by the Ultimate's firmware (3.15a), so a disk image or program directly in `/usb0/` or `/sd/` gets no automatic settings file, and the baseline file lives in a folder (`uboot64/uboot64.cfg`). See [Program settings files](#).

Credits

UBoot64 v2

Boot menu for C64 Ultimate enabled devices

Written in 2023 (v1), 2026 (v2) by Xander Mol <https://github.com/xahmol/UBoot64-v2> <https://www.idreamtin8bits.com/>

Inspired by and code used from DraBrowse: DraBrowse (db*) is a simple file browser. Originally created 2009 by Sascha Bader. Used version adapted by Dirk Jagdmann (doj) <https://github.com/doj/dracopy>

Most of code and functionality ported from: DMBoot 128: Device Manager Boot Menu for the Commodore 128 Written in 2020-2023 by Xander Mol <https://github.com/xahmol/DMBoot>

Built using:

- Oscar64 cross-compiler by drmortalwombat <https://github.com/drmortalwombat/oscar64>

Additionally uses code from:

- Ultimate 64/II+ Command Library Scott Hutter, Francesco Sblendorio <https://github.com/xlar54/ultimateii-dos-lib>
- ntp2ultimate by MaxPlap <https://github.com/MaxPlap/ntp2ultimate> Time via NTP code
- EPOCH-to-time-date-converter by sidsingh78 https://github.com/sidsingh78/EPOCH-to-time-date-converter/blob/master/epoch_conv.c
- petcom version 1.00 by Craig Bruce, 18-May-1995 Convert from PETSCII to ASCII, or vice-versa. https://codebase64.org/doku.php?id=source_conversion

Requires and made possible by the Ultimate II+ cartridge, Created by Gideon Zweijtzter <https://ultimate64.com/>

Bart van Leeuwen: For suggesting the default boot slot with configurable auto-boot timeout feature.

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